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Review
Future Microbiology

Brucella evasion of adaptive immunity

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The complex immune system of mammals is the result of evolutionary forces that include battles against pathogens, as sensing and defeating intruders is a prerequisite to host survival. On the other hand, microorganisms have evolved multiple mechanisms to evade both arms of immunity: the innate and the adaptive immune systems. The successful pathogenic intracellular bacterium *Brucella* is not an exception to the rule: *Brucella* displays mechanisms that allow evasion of immune surveillance in order to establish persistent infections in mammals. In this review, we highlight some key mechanisms that pathogenic *Brucella* use to evade the adaptive immune system.

Brucella: a stealthy bacterium at the heart of the innate immune system

Brucella & brucellosis

Brucella species are facultative Gram-negative intracellular bacterial pathogens responsible for brucellosis, a new emerging zoonosis. In natural hosts, the bacterium targets the reproductive tract and causes abortion and infertility [1]. However, the *Brucella* species that are the most pathogenic for humans are: *Brucella melitensis* (natural hosts: sheep and goats), *Brucella abortus* (natural hosts: cows) and *Brucella suis* (natural hosts: pigs). They are also relevant species from an economical and animal health perspective. In low-income countries, brucellosis is a public health problem [2,3]. Humans acquire *Brucella* from direct contact with infected animals or ingestion of infected dairy products. Human brucellosis is a debilitating febrile illness that can progress into a long-lasting disease with severe complications [2,4]. Many similarities exist between human and animal brucellosis, such as persistence in tissues of the mononuclear phagocyte system, including spleen, liver, lymph nodes and bone marrow. Furthermore, both in humans and animals, bacteria may be encountered within the bones and joints, as well as in the male reproductive organs and in the placenta and infected fetus [2,3]. If untreated, brucellosis becomes chronic, and in endemic areas (North, Central and South America, the Middle East, Mediterranean countries, north Africa, and countries of the Caucasus and central Asia), represents a serious health problem resulting in significant morbidity and economic loss [5].

Brucellosis is one of the most frequently encountered zoonoses worldwide, with 500,000 new cases diagnosed each year [6]. Although some vaccines are used in animals, no efficient and safe vaccine is yet available for humans. As human brucellosis is generally not transmitted from person to person (only two cases have been reported), the incidence of human infection is directly linked to the occurrence of brucellosis in a natural reservoir [4,7]. Thus, the identification of wildlife hosts that might compromise the effectiveness of control programs in livestock [8], along with the identification of new *Brucella* species that may represent a threat to humans, requires constant reappraisal of brucellosis symptoms [5].

It is difficult to study brucellosis in natural hosts due to practical, economical and ethical reasons. As a consequence, mouse models are widely used and have provided important mechanistic insights into the immune response to brucellosis. The course of murine brucellosis depends upon different bacterial (strain virulence, dose and inoculation route) and host parameters (genetic background, age, sex and physiological status) [9]. Three infection steps can be identified in mice infected with virulent *Brucella*: the first step occurs within 2 days of infection and corresponds to host invasion by the bacterium; the second phase is the acute phase (from 2 to 3 weeks) when bacteria replicate in different organs of the reticuloendothelial and reproductive systems; the last stage is the chronic phase (up to 6 months–1 year or more), displaying organ pathology [2,9,10]. The chronic phase starts with a steady chronic phase, during which the number of bacteria in the spleen and liver reaches a plateau (from 7 to 12 weeks), followed by the declining chronic phase, characterized by lower bacteria counts and diminished inflammation in the spleen and liver [2,9,10].

Keywords
■ adaptive immunity ■ *Brucella* ■ host–pathogen interaction ■ innate immunity ■ zoonosis

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Brucella pathogenicity

The main aspect of *Brucella* pathogenesis relies on its survival, replication and persistence within professional phagocytes, including macrophages and dendritic cells (DCs), as well as placental trophoblasts [11–13]. The bacterium has developed several strategies to resist and escape the killing mechanisms of phagocytic cells, as well as to interfere with the initiation of host immune responses.

Brucella controls the intracellular trafficking of its vacuole, named the *Brucella*-containing vacuole (BCV), in order to avoid degradation within phagolysosomes. BCVs then fuse with the endoplasmic reticulum (ER) membrane elements, including ER exit sites, thus ensuring a safe niche for replication [14,15]. The type IV secretion system (encoded by the *virB* operon), the cyclic β-glucan [16], the *Brucella* lipopolysaccharide (LPS) and the two-component regulatory system BvrS/BvrR [17] are among the key players for ensuring the intracellular life of *Brucella*. Recently, the transporter-like protein BacA, the flagellum-like structure and phosphatidylcholine were added to list of *Brucella* molecules that are required for survival within host cells [18]. Reaching the ER provides *Brucella* with a safe haven. Not only does the ER protect bacteria from humoral immunity, but it also keeps bacteria away from the potent bactericidal activity of phagocytic cells.

The physiological adaptation of *Brucella* to changing environments is crucial for its virulence [18,19]. It is also known that the in-built characteristics of the *Brucella* membrane envelope allow it to resist the humoral and cellular bactericidal activities of the host [20,21].

Brucella modulates the interplay between innate & adaptive immunity

As successful pathogens, *Brucella* spp. have evolved efficient strategies that enable them to circumvent and interfere with innate immune detection and therefore ensure the establishment of a chronic infection. However, reaching a chronic state of infection implies a balance between pathogen virulence and host resistance [22]. Although described as a stealthy pathogen, *Brucella* cannot be totally invisible to the immune system, which can clearly detect bacteria and develop immune responses.

The immune response to *Brucella* infection has mainly been characterized in mice. Similar to humans, *Brucella* spp. colonize the murine mononuclear phagocyte system and, in the absence of antibiotic treatment, can persist for months in tissues without causing significant morbidity or mortality [3]. During the acute phase of infection, *Brucella* actively replicates inside the macrophages and DCs of the target organs, such as the spleen, liver and lymph nodes. These organs display mononuclear inflammation and display the signs of pathological lesions, such as granuloma formation and extramedullary hematopoiesis [23]. Although innate immunity is capable of controlling *Brucella* replication during the acute phase in mice, efficient adaptive immunity is necessary to mount a strong immune response at later stages of infection [24].

Impressive & redundant mechanisms to evade innate immunity

The innate immune system is critical in the early control of bacterial replication and successful eradication of an infection. It is also linked to the adaptive immune response, which helps with the clearance of the infection and builds specific immunity with a memory component. Nevertheless, various successful pathogens have evolved multiple mechanisms to overcome host defense, including both innate and acquired immunity [25,26]. As a successful pathogen, *Brucella* has adopted a stealthy approach to cope with the innate immune system, avoiding sustained recognition by pattern recognition receptors (PRRs) and subsequent strong inflammatory responses [2,27]. Thus, the ultimate roles of the innate immune system in *Brucella* infection are to control bacteria replication and to provide the environment for generating strong Th1 host immune responses.

By targeting macrophages and DCs, *Brucella* runs the risk of activating an immune response that could lead to its elimination. Therefore, the structural features of *Brucella* allow bacteria to hide from the innate immune system. Indeed, *Brucella* spp. are devoid of conspicuous molecular determinants such as pili, fimbriae and capsules, and *Brucella* spp. possess noncanonical surface molecules, namely LPS, ornithine-containing lipids, lipoproteins and flagella, which are very weak inducers of innate immunity. These structures lack marked pathogen-associated molecular pattern activities, thus escaping recognition by PRRs to some extent [28]. Moreover, active mechanisms, such as interference with Toll-like receptor (TLR) signaling, contribute to the stealthy nature of *Brucella* at the onset of infection [29]. Escaping strong recognition by the innate immune system underlies the ability of *Brucella* to avoid full activation of infected macrophages and to maintain infected DCs

in an intermediate maturation stage [13,30]. Consequently, in the early stages (within the first 2 days) of *Brucella* infection in mice, the absence of strong inflammatory responses reflects an immunological delay of the host reaction. This gap is likely to provide the bacteria with a period of time within which they will reach the ER, protecting them from components of the humoral immune system and also keeping them away from the intracellular endocytic bactericidal mechanisms. It has been shown that *Brucella* localizes within the ER before the host macrophages or DCs are fully armed to fight the bacteria [15]. Reaching an intracellular safe haven within professional phagocytic cells nevertheless requires some mechanisms to avoid destruction. In addition to lowering host cell activation due to the lack of components with marked pathogen-associated molecular pattern activities, *Brucella* controls the fate of its BCV to reach a replicative niche, is resistant to harsh environments and displays metabolic versatility [2,3].

What advantages can be gained by residing within host immune cells? Maintaining an intracellular lifestyle protects *Brucella* from humoral immunity, but targeting professional immune cells might be seen as risky. However, invasion of DCs and macrophages could provide a means for bacteria to facilitate their spreading within the host. Moreover, it provides an opportunity for *Brucella* to subvert immune responses 'from the inside' by controlling the physiology of infected cells.

Brucella modulates the adaptive immune response

DCs play a key role in initiating and controlling the magnitude and the quality of adaptive immune responses [31]. However, several bacterial pathogens have developed mechanisms to subvert DC function and evade the immune system. *Brucella* has evolved mechanisms to interfere with information transmission from the innate to the adaptive immune system in order to evade the host immune response. The bacterium was recently found to efficiently proliferate within DCs both *in vitro* [13] and *in vivo* [12]. Despite extensive proliferation, *B. abortus* 2308 and *B. suis* 1330 have been shown to hinder DC phenotypic maturation *in vitro* [13,30]. Indeed, *Brucella*-infected DCs are characterized by low expression of MHC class II and CD80 and CD86 costimulatory molecules. This leads to inhibition of functional maturation, characterized by the absence of secretion of proinflammatory cytokines such as TNF-α and IL-12 and by inefficient antigen presentation to naive T cells [13,30].

Adaptive immune responses are critical for providing memory, a requisite property in long-lasting vaccination. IFN-γ-mediated type I immune responses are essential for the clearance of brucellosis [32]. The adaptive immune response in brucellosis can be classified into three mechanisms. First, IFN-γ produced by $CD4^+$, $CD8^+$, and γδ T cells activates the bactericidal function in macrophages in order to hamper the intracellular survival of *Brucella*. Second, cytotoxicity displayed by $CD8^+$ and γδ T cells kills infected macrophages. Third, Th1-type antibody isotypes, such as IgG2a and IgG3, opsonize the pathogen to facilitate phagocytosis towards degradative endocytic compartments. For instance, many serum passive transfer experiments suggest the significance of humoral immunity in murine brucellosis [2,32]. Moreover, cytokines play critical roles in immune responses, such as mediating innate and adaptive immunity and directing the immune response among immune-associated cells. The involved cytokines, regarded as key players in brucellosis, are IL-12, IFN-γ and TNF-α. However, much about these adaptive responses, in particular T-cell responses, remains to be discovered and characterized [2,6].

A striking feature during brucellosis is the ability of bacteria to survive within the host – the ultimate goal is the establishment of a chronic infection. Therefore, tight immune regulation is required during brucellosis. Immunosuppression is indeed detected in chronic brucellosis, with an increase of $CD4^+CD25^+$ T cells in the spleen, which in turn may play a role in the regulation of effector T lymphocytes [33]. Indeed, antibody-mediated depletion of murine $CD4^+$ or $CD25^+$ T cells induces *Brucella* elimination from target organs [33], and $CD4^+$ and MHC class IIAβ-deficient mutant mice are prone to *Brucella* elimination compared with wild-type mice. Immunosuppression may cause a deficient clearance of bacteria from infected spleens, livers and lymph nodes [34] due to the impaired activation of $CD8^+$ T lymphocytes and by the progressive decrease of macrophages and DC recruitment [35].

In addition, *Brucella* counteracts the development of a protective Th1 immune response by impairing the secretion of IL-12 and by inhibiting the T-cell stimulatory activity of infected DCs [13]. As a consequence, *Brucella*-infected DCs enter into a semi-mature phenotype characterized by low expression of MHC class II

and costimulatory molecules at the cell surface, a condition that is required to present exogenous protein antigens to specific T cells [30]. It is possible that, by preventing their full activation, *Brucella* uses the tolerogenic properties of DCs to subvert immune responses. There is growing evidence that induction of tolerance is not restricted to immature DCs. Recent studies have shown that migrating DCs are capable of inducing tolerance [36]. *Brucella*-infected DCs, which show an intermediate level of maturation, could therefore contribute to tolerance induction in tissues and to the establishment of a long-lasting chronic infection. Moreover, immature DCs interacting with naive $CD4^+$ T cells may induce the differentiation of Tregs and therefore inhibit a Th1 response in a TGF-β-dependent manner. This event has already been recorded in human chronic brucellosis [37]. Consistent with this hypothesis, reduction of IL-10 levels in mice has been shown to improve host resistance to *Brucella* infection [38]. This indicates that *Brucella,* by hampering antigen presentation, influences the interplay between innate and adaptive immunity.

Brucella LPS displays an unconventional structure that differs from enterobacterial LPS. *Brucella* lipid A is composed of a diaminoglucose disaccharide substituted with C16, C18, C28 and very long acyl chains that are structurally different from the canonical lipid A recognized by the TLR4–MD2 complex [39]. This nonclassical LPS structure seems to play a key role in the stealthy behavior of this pathogen, since *Brucella* LPS is poorly endotoxic [40]. These properties also impact on the initiation of adaptive immunity, since the presentation of antigenic peptides to T cells is affected [41]. Indeed, in peritoneal macrophages, purified *Brucella* LPS inhibits ovalbumin and hen egg lysozyme antigenic peptide presentation by MHC class II molecules to specific T-cell hybridomas. This is explained by the fact that *Brucella* LPS sequesters MHC class II molecules in large nonfunctional aggregates at the plasma membrane of macrophages. The biogenesis of these aggregates occurs in lysosomes [41]. Similarly, *Brucella* LPS also accumulates in MHC class II compartments in B lymphocytes [42]. As a consequence, the LPS is recycled from the antigen-loading and -processing compartments to the macrophage cell surface, forming large macrodomain clusters [43]. These structures are comprised of a mixture of lipid rafts, MHC class II molecules and *Brucella* LPS (Figure 1). Thus, the sequestration of an important proportion of MHC class II peptide complexes in macrodomains might influence the initiation of the immune response [43]. In correlation with this, it has been described that *Brucella* LPS-treated B cells and macrophages do not efficiently present antigens to $CD4^+$ lymphocytes [2,41,42]. However, neither the deficient uptake of the native antigen, nor reduced MHC class II surface expression, nor the binding of specific peptides to MHC class II molecules were involved in this event (Figure 1). It has been suggested that the deficient antigen presentation could be due to the inability of costimulatory molecules on the surface of antigen-presenting cells to make adequate contact for cell-to-cell recognition [42].

In conclusion, displaying an unconventional structure of LPS might be one of several strategies used by bacteria to escape the immune system and may explain in part the relative suppression of cell-mediated immunity observed in individuals with chronic brucellosis.

Brucella effector molecules control signaling events involved in DC maturation, which has consequences on antigen presentation and on T-lymphocyte activation. A gene encoding a TIR domain containing-protein named Btp1 or TcpB has been identified in *B. abortus* 2308 and *B. melitensis* 16M genomes. Due to the importance of this domain in TLR signaling, several studies have investigated the role of this protein in the context of DC maturation (Figure 1) [13,44]. It has been shown that *btp1*-mutant strains of *B. abortus* 2308 replicate normally in murine DCs, but partially enhance their phenotypic maturation and also increase the secretion of TNF-α and IL-12 *in vitro* compared with its parental wild-type strain [13]. During infection by live bacteria, TLR2, TLR4 and TLR9 are important for sensing infection. As discussed above, Btp1 inhibits DC maturation and the production of proinflammatory cytokines. This inhibition is the result of the ability of Btp1 to specifically interfere with TLR2 and TLR4 signaling [3,13]. It has been shown that Btp1 binds phosphoinositides and possibly colocalizes with MAL (also known as TIRAP) in the cell membrane [45]. This would allow Btp1 to elicit the targeted degradation of the signal-competent phosphorylated form of MAL, thereby allowing Btp1 to specifically interfere with TLR2 and TLR4 signaling [29]. Thus, *Brucella* interferes with TLR-dependent pathways in infected DCs in order to impair their function. How does Btp1 mediate MAL degradation? This remains to be uncovered. The recently described association of Btp1 with host microtubules, protecting them from depolymerization,

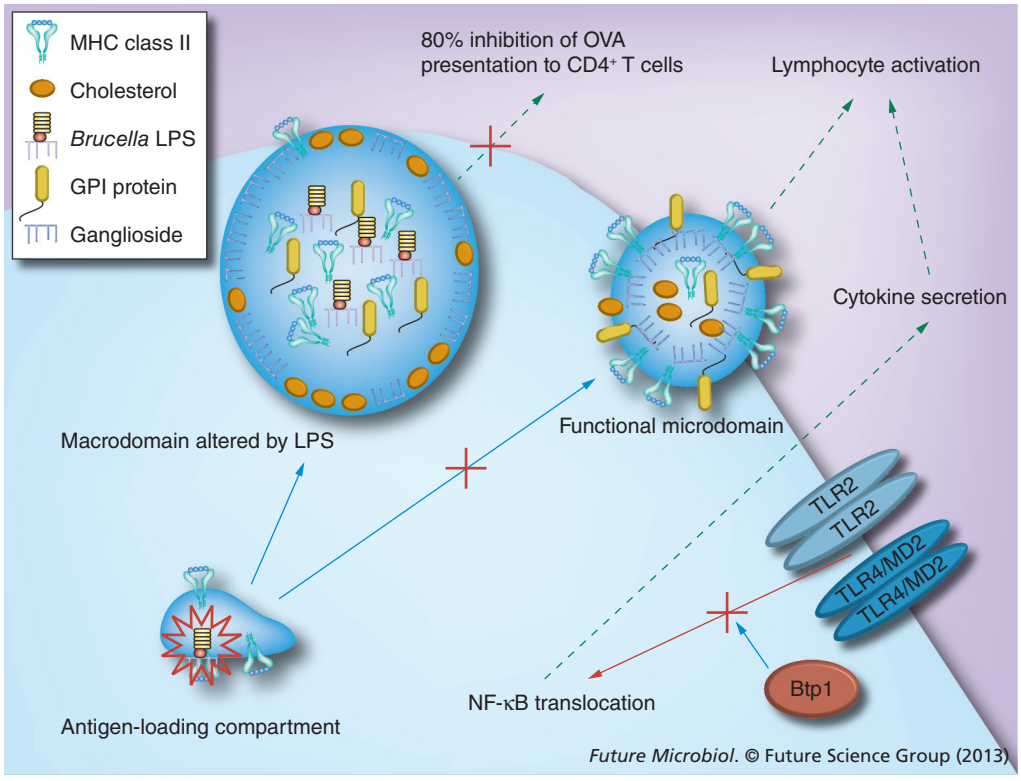


Figure 1. Two actions of *Brucella* on immunity. At steady state, the biogenesis of functional microdomains in macrophages is essential for antigen presentation to CD4+ T cells. These are formed in lysosomal compartments responsible for peptide loading to MHC class II molecules. *Brucella* LPS is capable of interacting with MHC class II molecules in lysosomes for more than 24 h and induces the generation of large macrodomains. These structures are then targeted to the cell surface and sequester MHC class II molecules. The macrodomains remain for a long period of time, even after bacterial death. The consequence is the impairment of antigen presentation. On the other hand, the *Brucella* Btp1 molecule interferes with the TLR2 and the TLR4/MD2 signaling pathways, which in turn induces inhibition of proinflammatory cytokine secretion and lymphocyte activation. Dashed arrows are related to adaptive immunity pathways and nondashed arrows are related to innate immunity pathways. Red star: LPS concentration and aggregation occur in the antigen-loading compartment.
GPI: Glycosylphosphatidylinositol; LPS: Lipopolysaccharide; OVA: Ovalbumin; TLR: Toll-like receptor.

could reflect another function for Btp1 that has been demonstrated *in vivo* [46].

Furthermore, it has recently been shown that Btp1 represents a novel effector of adaptive immune evasion by inhibiting CD8+ T-cell killing of *Brucella* epitope-specific target cells in mice [6]. This inhibition would result from the binding of Btp1 to phosphatidylinositol-4,5-bisphosphate, either blocking its action or sequestering it away from the antigen-presenting cell side of the immunological synapse [6]. Thus, the discovery of the new effectors that are able to interfere with TLR signaling strengthen the notion that *Brucella* has shaped an evolutionary trait to evade the adaptive immune system [28].

Future perspective

The innate immune system relies on PRRs diversely localized within the host to accurately detect infection caused by invading pathogens and subsequently trigger an efficient immune response. Nevertheless, various successful pathogens have evolved multiple mechanisms to overcome or alter many normally very effective host defense mechanisms, including both innate and acquired immunity. The intracellular pathogen *Brucella* is not an exception to the rule in displaying mechanisms that enable its evasion of immune surveillance and that are required to establish persistent infections in mammals. However, the host is capable of detecting *Brucella* and shaping a tailored immune response to infection. This leads to a chronic state of infection characterized by a balance between *Brucella* virulence and host resistance.

From the host point of view, much remains to be learned about the ability and the necessity of *Brucella* to control the immune response. For example, the ability of *Brucella* to regulate DC maturation *in vivo* has yet to be investigated.

Reducing the process of DC maturation, including the secretion of the TG1-prone cytokine IL-12, as well as antigen presentation, would enable *Brucella* to limit the development of the protective cellular-mediated immunity required against intracellular bacterial infections, in order to promote the establishment of chronic infections. Nevertheless, while *Brucella* has been shown to target DCs during infection, its ability to prevent their maturation has been so far restricted to *in vitro* observations. A recent *in vivo* study indicated that, upon *Brucella* infection, none of $CD11c^+$ mature cells migrating to the T-cell area were infected by *B. melitensis*, suggesting that infection of DCs could impair their migration and maybe their maturation [11]. Although these results are in agreement with the stealthy strategy attributed to *Brucella* and with the *in vitro* studies [13,30], the relevance of blocking infected DC maturation *in vivo* has yet to be established.

The study of the immune regulation events during brucellosis is one of the biggest challenges in the field. Several studies have suggested that the impaired activation of effector T lymphocytes and the progressive decrease of macrophages and DC activation might be due to regulatory events [35]. This could explain the deficient clearance of bacteria from the target organs [34]. Moreover, the ability of *Brucella* to induce an intermediate level of DC maturation would possibly enable it to use the tolerogenic properties of these cells in order to maintain chronic infections [13]. A recent study has demonstrated that the establishment of the high plateau phase of infection is dependent on nonantibody-mediated B-cell effector mechanisms, including B-regulatory functions, during murine brucellosis [32]. However, the connection between the suppressive effects of B cells on effector T cells has yet to be established.

Future years will unravel new *Brucella* effectors. In combination with a complex multiparametric analysis of the different immune cells involved in brucellosis, this may allow us to propose a better and more detailed assessment of protective immunity and therefore help to design efficient vaccines.

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Executive summary

Is Brucella *a stealthy pathogen?*

- *Brucella* and human brucellosis: humans acquire *Brucella* from direct contact with infected animals or ingestion of infected dairy products. Human brucellosis is a debilitating febrile illness that can progress into a long-lasting disease with severe complications. Brucellosis is one of the most frequently encountered zoonoses worldwide, with 500,000 new cases diagnosed each year.
- *Brucella* pathogenesis: the *Brucella*-containing vacuole fuses with the endoplasmic reticulum membrane ensuring a safe niche for replication within macrophages and dendritic cells. The type IV secretion system (encoded by the *virB* operon), cyclic β-glucan, lipopolysaccharide (LPS) and the two-component regulatory system BvrS/BvrR are among the key players for ensuring the continued intracellular life of *Brucella*.

Brucella *modulates the interplay between innate & adaptive immunity*

- Mechanisms to evade innate immunity: *Brucella* spp. are devoid of molecular determinants that alert the innate immune system, such as pili, fimbriae and capsules. *Brucella* expresses noncanonical surface molecules, namely LPS, ornithine-containing lipids, lipoproteins and flagella, which are very weak inducers of innate immunity.
- *Brucella* modulates the adaptive immune response: upon initation of the chronic phase of brucellosis, an increase of $CD4^+CD25^+$ T cells in the spleen is observed. This subset of lymphocytes might be involved in the regulation of effector T cells, thus leading to the establishment of an immunosuppressive state. Accordingly, *Brucella* effector proteins target Toll-like receptor (TLR) signaling in order to dampen T-cell stimulatory activity of infected dendritic cells and interfere with the development of a protective Th1 immune response. *Brucella abortus* LPS induces the formation of macrodomains sequestering MHC class II molecules rendered inactive for $CD4^+$ T-cell activation. Btp1 specifically interferes with TLR2 and TLR4 signaling, preventing T-cell activation.

Conclusion & future perspective

- *Brucella* is detected by the host at the onset of infection, which shapes a tailored response to infection. However, in many cases, this leads to a chronic phase controlled by *Brucella*. The discovery of mechanisms involved in chronicity is a challenge.
- The ability of *Brucella* to regulate dendritic cell maturation *in vivo* has yet to be investigated.
- The roles of the different immune cell players need to be investigated *in vivo*.
- The discovery of *Brucella* effector proteins in the control of the host immune response is a challenge for the next few years.

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